Thiago da Mata

Big Data Software Engineer

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Australian Citizen

Summary

I am a software engineer and technology leader with 20+ years of experience, having held roles ranging from developer to CTO. I thrive in roles that require creative problem-solving and autonomy, seeking opportunities to gain valuable skills while leveraging my knowledge and capabilities to deliver impactful, useful, reliable, elegant, clear, and cost-effective products.

I bring deep expertise in distributed systems, cloud computing, real-time processing, and big data pipelines, and I design and deliver mission-critical systems with high reliability and performance. My work has been recognised with three Cannes awards, reflecting a consistent track record of innovation and impact. I apply analytical rigour and careful evaluation, underpinned by a Master's in Biomedical Engineering with published research, to ensure systems are robust, maintainable, and built to last.

Committed to excellence, I mentor teams and guide projects from concept to production, combining leadership, technical mastery, and strategic insight to deliver high-value results. I focus on turning complex challenges into reliable, scalable solutions, balancing product demands with long-term vision to create systems that truly make a difference.

This document is the comprehensive version of my CV, intended for readers who prefer full detail. For a streamlined version, please see the one-page CV here: https://thiagomata.com/pdf/thiago-mata-one-page-cv.pdf.

Education

Master's Degree, Biomedical Engineering

2011 - 2013

Universidade de Brasília - UnB, with four publications.

Graduate Degree, Computer Science

2002 - 2008

Universidade de Brasília - UnB.

Experience

Senior Manager - **Macquarie Bank**

2024 - Present

https://www.macquarie.com.au/
Sydney, Australia

Helping the Banking and Financial Services to provide fast and consistent responses related to the company's data products. That includes consuming events from multiple sources, combining real-time events and daily snapshots, and deriving properties into a single interface that provides a concise, rich overview of customers' accounts.

Git, Java, Kafka, CI/CD, Protobuf, gRPC, Google BigQuery, Apache Beam, Google Dataflow, Google Cloud Storage, Google Spanner, Google Cloud Functions, Google Pub/Sub, Elasticsearch, Dynatrace, Docker, Helm, Sumo Logs, and Kubernetes.

Senior Data Engineer - **ROKT**

2019 - 2024

https://rokt.com/
Sydney, Australia

I initially worked on the Customer Data Platform, a real-time distributed system for organising, cleaning, validating, and enriching customer data. Then, we built the Identity System to group customer interactions for better behaviour analysis and the Attribution System to link interactions with advertiser conversion events, providing key campaign success metrics for machine learning models. Later, on the Traffic Resilience team, I led the Bot Detection service, monitoring and safeguarding the company's traffic from bots through real-time and offline analysis and statistical inference. Our evaluations, including rate limiting, local residency enforcement, and anomaly detection, run in Kubernetes Edge Clusters with Istio, Envoy Proxy, and gRPC-integrated services. Finally, I worked in the Anomaly Detection service to monitor and detect unexpected changes in key metrics.

5 years and 5 months

Git, Java, Scala, Python, Go, Gremlin, Spark, Kafka, AWS SQS / SNS, AWS Redshift, AWS Neptune, AWS EC2, AWS EKS, AWS CloudFormation, AWS CloudWatch, AWS DynamoDB, Terraform, Elasticsearch, Redis, HBase, Grafana, Prometheus, DataDog, CICD, Buildkite, Docker, Helm, Kubernetes, Istio, Envoy and GRPC.

Senior Data Engineer - **SLI Systems**

2017 - 2019

https://www.sli-systems.com/
Christchurch, New Zealand

On the Analytics Team, we worked on adding new features to existing products, creating new products with high-volume data, prototyping, and conducting online and offline testing of new machine learning strategies using Cloud Computing. I translated the models from the Data Scientist into SQL and Spark Streaming, and provided metrics on the impact and cost of these new features. We also migrated many company services from Apache Spark to Apache Beam, improving performance and reducing costs.

1 year and 8 months

Git, Java, Python, Google BigQuery, Apache Beam, Google Dataflow, Google Cloud Storage, Google Spanner, Google Pub/Sub, Elasticsearch, CI/CD, Jenkins, Docker, Helm, and Kubernetes.

Chief Technology Officer - **Ideia Big Data**

2014 - 2017

http://ideiabigdata.com
Brasília, Federal District, Brazil

I defined and drove the company's strategy and vision in designing, modelling, and developing its product for research, microtargeting, communication, and statistics, utilising publicly available demographic data, distributed systems, machine learning, and cloud-based streaming. We had to deal with Data Cleaning, Data Mining, Feature Extraction and Data Visualisation. I also improved the company development process and training policy, defined the career ladder, reported the company strategy and vision to investors, improved the observability and monitoring, evaluated the existing and new features with clients, measured their engagement and feedback, and optimised the cloud infrastructure for scalability and reliability while reducing the cost.

3 years

Git, PHP, Python, R, Scala, Node.js, Postgres, Hadoop, Apache Spark, Redis, Memcached, Grafana, MongoDB, AWS EC2, AWS SQS / SNS, AWS S3, AWS Elastic Beanstalk, Cloudera, Elasticsearch, Vagrant, Docker, Angular, Laravel, Cordova, Bootstrap, JQuery, D3.js and PDF.js, Karma and Selenium.

Senior Programmer - **Adjetiva**

2014 - 2014

http://www.adjetiva.com.br/
Brasília, Federal District, Brazil

Here, I created the brand channel of Embratur, mobile applications for the World Cup Events, educational games, and the Fiat EcoDrive desktop application. We also developed the SAT-JF14, a mobile application that won 3 Lions at Cannes.

10 months

Java, Node.js, Angular 2, PHP, Laravel, Postgres, UML, Scrum, Cordova, Node Webkit and Phaser.

Co-Founder - **OtimiCar**

2013 - 2014
4 months

http://www.otimicar.com
Brasília, Federal District, Brazil

Otimicar is a hardware-and-software solution that converts any car into a smart car. In Otimicar, I worked with the first version of the server's specification and development. This server communicates with hardware devices and partners to synchronise car data and send service and maintenance recommendations to car owners. It also generated reports for partners about their customers and products.

Git, Java, Node.js, REST, SOAP, MongoDB, UML, BPMN, PostgreSQL, Google Web App, and IoT.

Senior Coordinator - **Grupo 954**

2012 - 2013

http://www.grupo954.com
Brasília, Federal District, Brazil

The Grupo 954 provided clients with many data analyses, reports, and prediction models, including General Motors, one of Brazil's most significant car companies. I have modelled the company's internal business processes for estimating, specification, and development. Most of all, I worked on developing web applications for the company's clients.

1 year and 6 months

Git, PHP, Oracle Database, MS SQL Server, Rest, UML, BPMN and Scrum.

Process Analyst - **INEP** http://www.inep.gov.br/ Brasília, Federal District, Brazil	*2009 - 2012*
Here, I worked on many software systems with high availability, high data volumes, and a high number of concurrent users, including the E-MEC, which is software for educational accreditation and quality assurance for all universities in the country, and Revalida, the National Examination for the Revalidation of Medical Diplomas Issued by Foreign Higher Education Institutions.	3 years and 1 month
PHP, Laravel, Zend, Java, JBoss Seam, Memcached, Oracle, Redis, BPMN and Scrum.	
Teacher - **Studio On-Line** http://www.studioonline.com.br/ Brasília, Federal District, Brazil	*2004 - 2010*
Here, I taught PHP, MySQL, HTML, CSS and JavaScript.	6 years
PHP, MySQL, HTML, CSS, JavaScript	
Analyst and Web Developer - **SECOM** http://www.secom.gov.br/ Brasília, Federal District, Brazil	2008-2009
I worked with a Survey of Requirements, Modelling, Prototyping and Web Development there.	7 months
PHP, Oracle Database, Postgres, Zend and JavaScript.	
Database Administrator - **Bancoob** http://www.bancoob.com.br/ Brasília, Federal District, Brazil	2007 - 2008
Here, I developed the system for portfolio credit transfers, handling a high volume of financial transactions, and ensured the accounting closing process. I utilised storage procedures, memory tables, transactions, and bulk loading to handle heavy data manipulation.	1 year and 2 months
Visual Basic, MS SQL Server, Java, Crystal Reports, and UML.	
Web Developer - **Ministério da Educação** http://www.mec.gov.br/ Brasília, Federal District, Brazil	2006 - 2007
We developed the System of Information and Supervision of Professional Education and Technology and the Integrated Planning, Budget and Finance System of Brazil's Ministry of Education. Those systems handle high volumes of real-time and batch data, providing fast, dynamic reports.	1 year
SVN, PHP, Zend, Postgres, JavaScript, and SVG.	
Senior Developer - **AgênciaClick - Isobar** https://www.isobar.com Brasília, Federal District, Brazil	2005 - 2006

Here, we developed the SISAC - System for Monitoring and Controlling Communication Actions to the Presidency of the Republic of Brazil, the BrazilTour site, and the RMA - Network of NGOs of the Atlantic Forest site.	1 year
PHP, Postgres, Oracle and JavaScript.	
Developer - **SAMU** Brasília, Federal District, Brazil	2005 - 2005
The SAMU is a critical call centre software for emergency calls that connects the patient, ambulance driver, and medical team in one call. The project enables the emergency service to understand the patient's situation accurately and to define the appropriate team, medication, and nearby hospital for each case. There, I worked as a front-end Software Developer to develop the Internal System for the Ambulance Request Manager.	6 months
Java, JSTL, CSS, JavaScript, Struts, and XUL.	
Developer - **M.S.D.** Brasília, Federal District, Brazil	2003 - 2004
Here, we developed some computer-based training using Adobe Authorware, Adobe Flash, JavaScript and ToolBook.	11 months
JavaScript, ToolBook, Adobe Authorware, Computer-Based Training and Adobe Flash.	

Knowledge and Skills

PHP	*JavaScript*	*Python*	*Node.js*
Java	*Scala*	*Go*	*Visual Basic*
C#	*Dafny*	*Haskell*	*R*
Spark	*Hadoop*	*Apache Beam*	*Kafka*
MsSQL Server	*Postgres*	*Oracle Database*	*MySQL*
MongoDB	*Redis*	*ElasticSearch*	*HBase*
AWS EMR	*AWS Athena*	*AWS DynamoDB*	*AWS Redshift*
AWS Elastic Beanstalk	*AWS Lambda*	*AWS SQS/SNS*	*AWS EC2*
AWS CloudFormation	*AWS CloudWatch*	*AWS CW Logs*	*AWS Neptune*
Google Big Table	*Google Spanner*	*Oracle Cloud*	*Terraform*
Jenkins	*BuildKite*	*Git*	*Bash*
Kubernetes	*Docker*	*Istio*	*Envoy*

Languages

English *71/100 PTE* Portuguese *Native*

Certifications

Parallel programming 2024
École Polytechnique Fédérale de Lausanne

Programming with Google Go 2023
University of California, Irvine

Solutions Architect Associate 2022
Amazon Web Services

Functional Program Design in Scala 2021
École Polytechnique Fédérale de Lausanne

Introduction to Graph Databases, Cypher, and Neo4j 2020
Pluralsight

Practical Machine Learning 2019
Johns Hopkins University

AWS Fundamentals: Going Cloud-Native 2019
Amazon Web Services

Regression Models 2019
Johns Hopkins University

Deep Learning in Python Course 2017
DataCamp

Leveraging Unstructured Data with Cloud Dataproc 2017
Google Cloud Platform

Big Data and Machine Learning Fundamentals 2017
Google Cloud Platform

Apache Spark 2.0 + Scala: DO Big Data Analytics & ML 2017
Udemy - V2 Maestros

Reproducible Research 2017
Johns Hopkins University

Functional Programming Principles in Scala 2016

École Polytechnique Fédérale de Lausanne

Exploratory Data Analysis 2016
Johns Hopkins University

Getting and Cleaning Data 2016
Johns Hopkins University

R Programming 2016
Johns Hopkins University

Introduction to Probability and Data 2016
Duke University

The Data Scientist's Toolbox 2016
Johns Hopkins University

Advanced Track of Introduction to Artificial Intelligence 2011
Stanford University

Publications

Critérios de Usabilidade e Acessibilidade para Avaliação de Aplicações Educacionais 3D na Web 2015
CBIE-LACLO 2015

Definition of Computational Criteria to Evaluation of 3D Anatomical Web Manikins for Teaching in Health Science and for Medical Applications 2013
PAHCE – Pan American Health Care Exchanges

Accessibility for Web 3D Anatomical Manikins 2013
SETI - First Symposium on Education and Inclusive Technology
Regional Integrated University of High Uruguay and Missions

Definition of criteria for evaluating 3D anatomical manikins on the web for education in healthcare or medical applications 2012
Extensio: Electronic Journal of Extension

Honours & Awards

Honourable Mention in Ranking of the Best Published Articles - 2015
Conferência Latino-Americana de Objetos e Tecnologias de Aprendizagem - LACLO

Gold Lion - 1º Place - Category Media, Entertainment and Leisure 2014
Cannes Lions - Campaign SAT-JF14. Record Collection Music Album - John Frusciante

Silver Lion - 2º Place - Category Use of Digital Platforms 2014
Cannes Lions - Campaign SAT-JF14. Record Collection Music Album - John Frusciante

Bronze Lion - 3º Place - Category New Product Launch 2014
Cannes Lions - Campaign SAT-JF14. Record Collection Music Album - John Frusciante